Mohamed Abuella

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Scholar: Mohamed Abuella

Summary

Electrical engineer by training, traditionally interested in mathematical and computational analysis, modeling, and optimization, and data analytics for integrated energy systems planning and operations.
As a researcher, I focus on modernizing the electric grid and optimizing the integration of distributed energy resources by applying descriptive, predictive, and prescriptive analytics to improve energy efficiency and support the transition to net-zero infrastructures. My work includes recent applications in hydrogen and maritime shipping, with a broader interest in utilizing Artificial Intelligence to foster sustainability. Seeking to transfer and expand my expertise to bridge the gap between advanced research and real-world application.
Keywords: *Energy systems modeling and analytics, planning and operation, renewable energy and hydrogen integration, optimization and game theory, energy transition policy, net-zero pathways, AI/ML for energy forecasting and efficiency.*
For more details, see my online CV and professional profile in mohamedabuella.github.io/cv.

Education

University of North Carolina at Charlotte (UNCC) **USA**
Ph.D in Electrical Engineering, GPA 4.0 *2014–2018*

Southern Illinois University at Carbondale (SIUC) **USA**
M.Sc in Electrical and Computer Engineering, GPA 4.0 *2010–2012*

Higher Polytechnic Institute & College of Industrial Technology at Misurata **Libya**
DipHE in Instrumentation 82%, and B.Tech Electromechanical Engineering, 86% *2002–2008*

Experience

Independent Researcher **UK**
PyPSA-Libya Project *August 2026–Present*
Developing PyPSA-Libya, an open-source energy system model based on the PyPSA-Earth framework to study renewable energy integration and sector-coupled optimization for national decarbonization pathways and hydrogen export opportunities in Libya.
For more details, see GitHub Repository and arXiv Preprint.
○ Acquired Expertise: *Open-Source Energy System Modeling, PyPSA, Renewable Energy Integration & Planning, Hydrogen Trading.*

Research Fellow **UK**
Northumbria University, Department of Mathematics, Physics and Engineering *May 2024–May 2026*
Supporting the delivery of the "Hydrogen Integration for Accelerated Energy Transitions (HI-ACT)" research project.
See more details in HI-ACT Project.
○ Acquired Expertise: *Integrated Energy Systems, Game Theory, Strategic & Policy Modeling, Hydrogen Integration, Energy Transition*

Researcher **Sweden**
Halmstad University *April 2022–April 2024*
Postdoctoral Researcher at the Center for Applied Intelligent Systems Research (CAISR). Research on AI for Sustainability applying Machine Learning techniques.
For more details in iHelm Project.
○ Acquired Expertise: *Integrated Industry-Academia Collaboration, Big-Data & Spatiotemporal Analysis, Explainable AI (XAI), Research Methodologies & Project Supervision.*

Lecturer **Libya**
College of Industrial Technology at Misurata *February 2020–March 2022*
Taught Electrical Circuits, Electrical Measurements, Math 101.
○ Acquired Expertise: *Curriculum Revision & Preparing, Dedication, Listening, "Try to Modeling the Student's Way of Thinking".*

Research Assistant **USA**
Energy Production and Infrastructure Center (EPIC) at UNC Charlotte *2014–2019*

Statistical and Predictive Analytics to Modernize the Grid and Optimize its Integration of Renewables, focusing on Solar Energy Resources.
For more details in Graduate Research Portfolio.
○ Acquired Expertise: *Energy Analytics, Energy Markets, Renewable Energy & Supply Chain, Machine Learning.*

M.Sc Research **USA**
Department of Electrical and Computer Engineering at SIUC *2010–2012*
Optimization for Electric Power Systems Including Wind Power.
For more details in Graduate Research Portfolio.
○ Acquired Expertise: *Power Systems Analysis, Operation and Planning, Systems Optimization, Smart Grid.*

Teaching Assistant and Lab Instructor **Libya**
College of Industrial Technology at Misurata *2008–2009*
Taught Mathematics, Power Systems Analysis, and PLC.
○ Acquired Expertise: *Teaching, Tutorials, Lab Modeling & Simulations.*

Electrical Technician **Libya**
Residential Electrical Wiring and Water & Wastewater Company *2000–2008*
Wiring and maintain electrical control equipment.
○ Acquired Expertise: *Electrical Wiring & Installations, Maintenance & Operation.*

Recognitions

Best Paper Award (Third Prize): RPG 2025 – 14th Int. Conf. on Renewable Power Generation, Shanghai, China *2025*

Outstanding Reviewer: IEEE Transactions on Sustainable Energy *2017*

Third Prize for Student Papers: The 47th North American Power Symposium *2015*

The Institute Prize for team of Energy Analytics Lab at UNCC: Global Energy Forecasting Competition *2014*

The 1st Place: Department of Electromechanical Engineering at College of Industrial Technology *2008*

Publications

Authored numerous peer-reviewed publications (all as a leading author). For a complete list of works, please see my Google Scholar profile: Mohamed Abuella (accumulating around 800 citations, h-index: 10) and my ORCID.

Journal Articles

1. M. Abuella, A. Allahham, S. L. Walker, "A Policy-Navigation Framework for Exploring Hydrogen Integration Pathways in Great Britain Towards Net Zero," *Energy Policy*, vol. 218, Article 115535, Nov. 2026. DOI: 10.1016/j.enpol.2026.115535.
2. M. Abuella, A. Allahham, S. L. Walker, "Game Theory Approaches to Hydrogen Infrastructure Investment Planning in Great Britain: A Comparative Analysis of Competitive and Cooperative Frameworks," *International Journal of Hydrogen Energy*, vol. 220, p. 154064, 2026. DOI: 10.1016/j.ijhydene.2026.154064.
3. M. Abuella, A. Allahham, S. L. Walker, "A Cooperative Planning Framework for Hydrogen Blending in Great Britain's Integrated Energy System," *Energies*, vol. 19, no. 9, p. 2018, 2026. DOI: 10.3390/en19092018.
4. M. Abuella, A. Allahham, N. A. Rufa'I, S. L. Walker, "Hydrogen-Based Decarbonisation Strategies for Residential Heating: An Energy Efficiency and Conservation Analysis in the North of Tyne Region," *Energies*, vol. 18, no. 23, p. 6237, 2025. DOI: 10.3390/en18236237.
5. M. Abuella, H. Fanaee, S. Nowaczyk, S. Johansson, E. Faghani, "Time-Series Analysis Approach for Improving Energy Efficiency of Fixed-Route Passenger Vessel in Short-Sea Shipping," *Ocean Engineering*, vol. 334, p. 121555, 2025. DOI: 10.1016/j.oceaneng.2025.121555.
6. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Spatial Clustering Approach for Vessel Path Identification," *IEEE Access*, vol. 12, pp. 66248–66258, 2024. DOI: 10.1109/ACCESS.2024.3399116.
7. M. Abuella and B. Chowdhury, "Forecasting of Solar Power Ramp Events: A Post-Processing Approach," *Renewable Energy*, vol. 133, pp. 1380–1392, 2019. DOI: 10.1016/j.renene.2018.09.005.
8. M. Abuella and B. Chowdhury, "Improving Combined Solar Power Forecasts Using Estimated Ramp Rates: Data-Driven Post-Processing Approach," *IET Renewable Power Generation*, vol. 12, no. 10, pp. 1127–1135, 2018. DOI: 10.1049/iet-rpg.2017.0447.

Working Papers

1. Abuella, M. (2026). *PyPSA-Libya: An Open-Source Energy System Model for National Decarbonization and Hydrogen Export Pathways*. arXiv preprint arXiv:2610.38094. DOI.

Conference Papers

1. M. Abuella, A. Allahham, S. L. Walker, "Socio-Technical, Electrification, and Hydrogen-Driven Pathways for Residential Heating Decarbonisation in the North of Tyne," in *9th International Conference on Environment Friendly Energies and Applications (EFEA 2025)*, 4–5 December 2025, Newcastle upon Tyne, UK.
2. M. Abuella, A. Allahham, S. L. Walker, "A Game-Theoretic Framework for Hydrogen Integration to Accelerate Energy Transitions in Great Britain," in *14th International Conference on Renewable Power Generation (RPG 2025)*, 24–26 October 2025, Shanghai, China.
3. M. Abuella, M. Atoui, S. Nowaczyk, S. Johansson, E. Faghani, "Data-Driven Explainable Artificial Intelligence for Energy Efficiency in Short-Sea Shipping," in *ECML PKDD*, 2023.
4. M. Abuella and B. Chowdhury, "Adjusting Post-Processing Approach for Very Short-Term Solar PV Power Forecasts," in *2021 IEEE MI-STA*, 2021.
5. M. Abuella and B. Chowdhury, "Qualifying Combined Solar Power Forecasts in Ramp Events' Perspective," in *IEEE Power and Energy Society General Meeting*, 2018.
6. M. Abuella and B. Chowdhury, "Forecasting Solar Power Ramp Events Using Machine Learning Classification Techniques," in *IEEE PEDG*, 2018.
7. M. Abuella and B. Chowdhury, "Hourly Probabilistic Forecasting of Solar Power," in *North American Power Symposium (NAPS)*, 2017.
8. M. Abuella and B. Chowdhury, "Random Forest Ensemble of Support Vector Regression Models for Solar Power Forecasting," in *IEEE ISGT*, 2017.
9. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Support Vector Regression," in *ASEM International Annual Conference*, 2016.
10. M. Abuella and B. Chowdhury, "Solar Power Forecasting Using Artificial Neural Networks," in *North American Power Symposium (NAPS)*, 2015.
11. M. Abuella and B. Chowdhury, "Solar Power Probabilistic Forecasting by Using Multiple Linear Regression Analysis," in *IEEE SoutheastCon*, 2015.